

Combinatorics – 2

Recurrences and generating functions

Homework

14 сентября 2026 года

Theoretical minimum. In the ring $\mathbb{F}[[x]]$ of formal power series the coefficients of a product are the convolution of the coefficients, a series is invertible exactly when its constant term is non-zero, and identities such as $(1-x)^{-1} = \sum x^n$ require no convergence; the analytic side is Functional sequences, series and spaces – 3. A sequence satisfies a linear recurrence $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ if and only if its generating function is $P(x)/Q(x)$ with $Q = 1 - c_1 x - \dots - c_k x^k$ and $\deg P < k$; partial fractions then give $a_n = \sum_i p_i(n) \lambda_i^n$ over the roots λ_i of the characteristic polynomial, the degree of p_i being less than the multiplicity. Disjoint union adds generating functions, pairing multiplies them, and an arbitrary sequence of pieces counted by F with $F(0) = 0$ is counted by $1/(1-F)$; for labelled objects the same rules hold for exponential generating functions, and a set of components counted by F is counted by e^F . Finally, objects that are walks in a finite digraph are counted by powers of its adjacency matrix, with rational generating function of denominator $\det(I - xM)$ and growth rate the spectral radius.

1. For a positive integer n let $p(n)$ be the number of ways to write n as a sum of positive integers, with $p(0) = 1$. Prove that $p(n) - p(n-1)$ is the number of ways to write n as a sum of integers each of which is at least 2. (IMC)
2. Prove that the generating function of the central binomial coefficients is $\sum_{n \geq 0} \binom{2n}{n} x^n = (1-4x)^{-1/2}$, working formally with the binomial series, and deduce $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$ both by coefficients and by a double count.

3. Exponential generating functions: prove that the derangement numbers have exponential generating function $e^{-x}/(1-x)$, that the number of involutions of $[n]$ has $e^{x+x^2/2}$, and that the Bell numbers have e^{e^x-1} . Deduce a recurrence for the involutions.
4. Count the binary words of length n avoiding the factor 11, and those avoiding 101, by transfer matrices; show that the two growth rates are different and compute both to three decimals.
5. A finite collection of digits 0 and 1 is written around a circle; an arc of length L consists of L consecutive digits. Assume that $|Z(w)-Z(w')| \leq 1$ for any two arcs w, w' of the same length, where Z counts the zeros. Describe all such collections. (Putnam)
6. A sequence of positive integers is defined by $a_1 = a_2 = 1$, $a_3 = 2$ and, for $n \geq 4$, by

$$a_n a_{n-2} a_{n-3} = a_{n-1}^2 a_{n-3} + a_{n-1} a_{n-2}^2.$$

Prove that $S = \sum_{n \geq 1} 1/a_n$ is irrational. (IMC 2025, shortlist)

7. Prove the reflection principle: the number of lattice paths from $(0, 0)$ to (a, b) with unit right and up steps that touch the line $y = x + 1$ equals the number of unrestricted paths to the reflected endpoint. Deduce the ballot problem and the formula for the Catalan numbers (the numbers themselves are the subject of Combinatorics – 3).
8. Let $f(m, n)$ denote the number of n -tuples $(x_1, \dots, x_n)$ of integers with $|x_1| + \dots + |x_n| \leq m$. Show that $f(m, n) = f(n, m)$, and find the generating function of f in two variables. (Putnam 2005)
9. Prove that the number of ways to tile a $2 \times n$ rectangle by dominoes is a Fibonacci number and that the number of ways to tile a $3 \times 2n$ rectangle satisfies $t_n = 4t_{n-1} - t_{n-2}$; find the generating function and the closed form of the second sequence.
10. Prove that a power series $\sum a_n x^n$ with integer coefficients is rational with denominator $Q(0) = 1$ if and only if the Hankel determinants

$\det(a_{i+j})_{i,j \leq N}$ vanish for all large N , and use this to decide whether the sequence $a_n = \binom{2n}{n}$ satisfies a linear recurrence with constant coefficients.